EXATHON 2026 HACKATHON ENTRY

SIGNALFIRE AI

Find the warning signs hidden across your data.

Category: AI-Powered BI, Anomaly Detection & Cross-Dataset Risk Discovery

Core Idea: Unifies fragmented data silos, detects multi-metric lead-lag cascades, and formulates evidence-backed investigation hypotheses.

Analytical Architecture: Native Exasol Personal integration via pyexasol with local in-process DuckDB fallback.

Demonstration Domains: Retail Logistics & Industrial MRO Equipment Reliability.

PROBLEM STATEMENT

The Cost of Monitoring in Departmental Silos

Why isolated dashboards fail to catch emerging systemic operational crises

- **Disjointed Data Silos:** Fulfillment delays, customer complaints, refunds, and machinery telemetry are reviewed in silos.
- **The Lead-Lag Blind Spot:** Incidents unfold as multi-day chains (Carrier Delay Day 50 -> Complaints Day 52 -> Returns Day 54).
- **Alert Fatigue & False Alarms:** Static >25% change thresholds trigger relentless false alarms during normal demand cycles.
- **Compounding Financial Impact:** By the time executive dashboards alert, irreparable customer churn and downtime have occurred.

PRODUCT ARCHITECTURE

Cross-Dataset Intelligence & Epistemic Rigor

Rigorous epistemic separation between facts, statistical associations, and hypotheses

- **Observed Facts:** Deterministic data points directly computed from verified database records (e.g. 2.78-day delay).
- **Statistical Associations:** Empirical Pearson ($r=0.89$) and Spearman correlation coefficients with automated lead-lag offsets.
- **Hypotheses & Alternative Explanations:** Structured explanations for human investigation; explicitly tracks competing confounders.
- **Core Epistemic Mandate:** Never presents statistical correlation as proven causation.
- **Targeted Action Playbooks:** Proactive remediation steps (carrier SLA audit, supplier buffer prioritization).

DOMAIN VALIDATION

Dual-Domain Operational Demonstrations

Proving multi-metric lead-lag discovery across Retail and Industrial MRO sectors

• **Retail Logistics (North Hub Breakdown):** Transit delays surge from 0.10d to 2.78d (+2,568%) -> Customer complaints surge +961% -> Refund disbursements jump +276% ($14,200 loss).

• **Industrial MRO (Plant Alpha Stamping Press):** RMS vibration velocity surges from 2.1 to 8.4 mm/s (ISO 10816-3 breach) -> Overdue PM backlog surges 3.8x -> Bearing stockout -> Catastrophic 46.5-hr stoppage ($280k loss).

• **Control Cohorts:** Contrasted against unaffected cohorts (South/East/West; Plant Beta/Gamma) to isolate localized root causes.

HIGH-PERFORMANCE ANALYTICS

Exasol Personal Analytical Integration

In-memory column-store analytical SQL with zero-downtime DuckDB fallback transparency

- **Official pyexasol Driver:** Native WebSocket JSON-RPC connection executing high-throughput analytical rollups and aggregations.
- **Database Adapter Pattern:** Polymorphic abstraction completely isolates analytics from storage, switching between Exasol and DuckDB.
- **Strict Transparency Mandate:** Real-time database status indicator in top navigation bar; local calculations are NEVER labeled as Exasol-powered.
- **Zero-Friction Offline Evaluation:** Judges can evaluate the full application offline without Docker via DuckDB.

EMPIRICAL BENCHMARKS

Ground-Truth Evaluation & Accuracy Benchmark

Audited performance comparison against a traditional static percent-change baseline

- **SIGNALFIRE AI (Robust MAD & IQR):** Precision = 70.0% | Recall = 77.8% | F1-Score = 0.737 | False Positives = 3.
- **Traditional Baseline (>30% Change Rule):** Precision = 42.9% | Recall = 75.0% | F1-Score = 0.545 | False Positives = 8.
- **Key Advantage:** SignalFire's non-parametric Median Absolute Deviation fences filter cyclical demand noise while catching true multi-metric crises.
- **Automated Pytest Suite:** 12/12 unit and integration tests passing in 3.09s.

UI/UX & VISUAL ENGINEERING

Enterprise 3D Glassmorphism & Working 3D Effects

Tactile, refined interface interactions designed to enhance data comprehension

- **Interactive 3D Risk Universe:** Real-time HTML5 3D Canvas with drag-to-rotate pitch/yaw controls, auto-orbiting, and live telemetry HUD.
- **Real-Time 3D Gyro Tilt:** Dashboard cards physically tilt in 3D perspective based on mouse position with dynamic holographic specular light glare.
- **3D Modal Parallax:** Deep inspection modal card dynamically responds to mouse gyro coordinates with embossed tactile KPI tiles.
- **Custom 3D Glass Scrollbars:** Slim 7px gradient floating pill scrollbars across modal and table containers.

DECISION SUPPORT

Decision Support & What-If Sensitivity Simulator

Translating detected signals into verified conclusions and simulated financial risk

- **Dual-Domain What-If Simulator:** Interactive sensitivity sliders for Retail (delay %, return %, demand %, inventory %) and Industrial MRO (vibration %, PM backlog %, parts lead time, hourly cost).
- **Methodological Boundary:** Separates direct arithmetic calculations from model-based behavioral projections and stated engineering assumptions.
- **Investigation Workspace:** Decision-support ticket manager allowing analysts to assign operational priorities (P1-P3) and record audit conclusions.
- **Tool-Grounded AI Assistant:** Conversational analyst retrieving verified evidence from database queries, preventing hallucinations.

MARKET POSITIONING

Competitive Advantage: Beyond Siloed BI

Why traditional enterprise dashboards leave millions in unaddressed operational risk

- **Traditional BI (Tableau/PowerBI):** Sits on static data, requires manual cross-filtering, lacks automated lead-lag discovery and epistemic rigor.
- **APM / Monitoring (Datadog/PagerDuty):** Focuses on infrastructure uptime, relies on noisy static thresholds, misses cross-table business cascades.
- **SIGNALFIRE AI Advantage:** Automated multi-dataset correlation ($r=0.89$), noise-robust MAD normalization, control cohort benchmarking, and what-if simulation in one unified platform.

CONCLUSION & ROADMAP

Delivering Operational Resilience

Transforming hidden warning signs into proactive business interventions

- **What We Built & Verified:** Complete full-stack system, 13 datasets (60,653+ records), interactive 3D universe, 3D glassmorphism, 12/12 passing tests.
- **Near-Term Horizon:** Multi-node Exasol cluster deployment for multi-billion record datasets.
- **Multimodal Signal Ingestion:** Embedding vectors for customer call audio transcripts and technician maintenance notes.
- **Automated ERP Enforcement:** SAP and NetSuite webhook triggers for automatic vendor SLA penalty clawbacks upon confirmed lead-lag breach.